Unit B Review

Multiple Choice Bank — 75 Questions

Question Set

MC 1. According to the lesson material, what fundamentally drives oxidation-reduction reactions?

A. Electron transfer between atoms or ions
B. Proton transfer between molecules
C. Neutron exchange in the nucleus
D. Energy absorption from surroundings

Correct answer: A

Rationale: The material explicitly states that oxidation-reduction reactions are fundamentally about electron transfer between atoms or ions, where valence electrons move to achieve more stable configurations.

MC 2. When sodium reacts with chlorine to form sodium chloride, what happens to the sodium atom?

A. It gains one electron and becomes reduced
B. It loses one electron and becomes oxidized
C. It shares electrons equally with chlorine
D. It remains neutral throughout the reaction

Correct answer: B

Rationale: The material states that each sodium atom loses one electron to form Na^+ ions during NaCl formation, which is oxidation (loss of electrons).

MC 3. What is the oxidation number of carbon in CH_4?

A. +4
B. 0
C. -4
D. +2

Correct answer: C

Rationale: According to the material, in CH_4, hydrogen is +1, and since the compound is neutral, carbon must be -4 to make the sum zero.

MC 4. In the combustion of methane, carbon's oxidation number changes from -4 to +4. How many electrons does carbon lose per molecule?

A. 2 electrons
B. 4 electrons
C. 6 electrons
D. 8 electrons

Correct answer: D

Rationale: The material explicitly states that when carbon's oxidation number increases from -4 to +4 in methane combustion, carbon loses 8 electrons per molecule.

MC 5. What is the oxidation number of elements in their free state, such as Na or Cl_2?

A. +1
B. -1
C. 0
D. Varies by element

Correct answer: C

Rationale: The material states that elements in their free state always have oxidation numbers of zero, including Na, Cl_2, and O_2.

MC 6. According to the systematic rules, what is the oxidation number of hydrogen in metal hydrides?

A. +1
B. 0
C. -1
D. +2

Correct answer: C

Rationale: The material specifies that hydrogen has an oxidation number of +1 except in metal hydrides where it's -1.

MC 7. Why does sodium have a low ionization energy of 496 kJ/mol?

A. Its nucleus has many protons attracting the electron
B. Its single valence electron is far from the nucleus and shielded by inner electrons
C. It forms strong metallic bonds with other sodium atoms
D. Its electrons are in low-energy orbitals

Correct answer: B

Rationale: The material explains that sodium has low ionization energy because its single valence electron is far from the nucleus and shielded by inner electrons.

MC 8. What is the electron affinity of chlorine as mentioned in the material?

A. 496 kJ/mol
B. 787 kJ/mol
C. 349 kJ/mol
D. 100 kJ/mol

Correct answer: C

Rationale: The material specifically states that chlorine has a high electron affinity of 349 kJ/mol.

MC 9. In a neutral compound, what must the sum of all oxidation numbers equal?

A. +1
B. -1
C. The number of atoms
D. Zero

Correct answer: D

Rationale: The material states that the sum of all oxidation numbers in a neutral compound must equal zero.

MC 10. What is the oxidation number of oxygen in peroxides?

A. -2
B. -1
C. 0
D. +1

Correct answer: B

Rationale: According to the material, oxygen has an oxidation number of -2 except in peroxides where it's -1.

MC 11. Why do oxidation and reduction always occur together?

A. They require the same activation energy
B. Electrons cannot exist freely in solution
C. They both release energy

D. Chemical equations must balance

Correct answer: B

Rationale: The material explains that oxidation and reduction are coupled because electrons cannot exist freely in solution—when one species loses electrons, another must simultaneously gain them.

MC 12. What is the lattice energy mentioned for NaCl formation?

A. 349 kJ/mol
B. 496 kJ/mol
C. 787 kJ/mol
D. 1000 kJ/mol

Correct answer: C

Rationale: The material states that the energy released when Na^+ and Cl^- ions attract each other in the crystal lattice is 787 kJ/mol.

MC 13. According to the material, what configuration do chlorine atoms achieve by gaining electrons?

A. A duplet configuration
B. A stable octet configuration
C. An excited state
D. A metallic configuration

Correct answer: B

Rationale: The material states that adding an electron to chlorine completes its valence shell, creating a stable octet configuration.

MC 14. What is the oxidation number of Group 1 metals in compounds?

A. 0
B. +1
C. +2
D. -1

Correct answer: B

Rationale: The material provides the systematic rule that Group 1 metals have an oxidation number of +1 in compounds.

MC 15. In CO_2, what is the oxidation number of carbon?

A. -4
B. -2
C. +2
D. +4

Correct answer: D

Rationale: The material states that in CO_2, oxygen is -2, so carbon must be +4 to make the sum zero in this neutral compound.

MC 16. What type of bonds contain methane's high-energy electrons according to the material?

A. C=C bonds
B. C-H bonds
C. C=O bonds
D. O-H bonds

Correct answer: B

Rationale: The material states that methane's electrons are in high-energy C-H bonds.

MC 17. When an atom loses electrons, what happens to its oxidation state?

A. It decreases

B. It remains constant

C. It increases

D. It becomes zero

Correct answer: C

Rationale: The material defines oxidation as when an atom loses electrons, resulting in an increase in its oxidation state or oxidation number.

MC 18. What is the oxidation number of fluorine in compounds?

A. +1

B. 0

C. -1

D. -2

Correct answer: C

Rationale: According to the systematic rules provided, fluorine has an oxidation number of -1 in compounds.

MC 19. In the combustion of methane, what happens to oxygen?

A. It is oxidized from 0 to -2

B. It is reduced from 0 to -2

C. It is oxidized from -2 to 0

D. It remains unchanged

Correct answer: B

Rationale: The material states that in methane combustion, oxygen has oxidation number 0 in O_2, but -2 in the products, so oxygen is reduced (gains electrons).

MC 20. What drives the thermodynamic force for redox reactions?

A. The tendency of electrons to move from lower-energy to higher-energy orbitals

B. The tendency of electrons to move from higher-energy to lower-energy orbitals

C. The formation of covalent bonds only

D. The absorption of light energy

Correct answer: B

Rationale: The material states that the thermodynamic driving force for redox reactions comes from the tendency of electrons to move from higher-energy orbitals to lower-energy orbitals.

MC 21. What is the oxidation number of a monatomic ion?

A. Always zero

B. Always +1

C. Equal to its ionic charge

D. Opposite to its ionic charge

Correct answer: C

Rationale: The material states that for monatomic ions, the oxidation number equals the ionic charge (e.g., Na^+ = +1, Cl^- = -1).

MC 22. According to the material, what type of cells force electrons through external circuits to generate power?

A. Electrolytic cells

B. Voltaic cells

C. Concentration cells

D. Fuel cells

Correct answer: B

Rationale: The Structure → Function Reminder states that voltaic cells force electrons through external circuits to generate power.

MC 23. What determines the direction of spontaneous electron flow according to the material?

A. From higher to lower potential
B. From lower to higher potential
C. Always from anode to cathode
D. Always from positive to negative

Correct answer: B

Rationale: The Cause → Effect Summary states that electron transfer occurs spontaneously from lower to higher potential species.

MC 24. What type of cells use external power to force electrons against their thermodynamic preference?

A. Voltaic cells
B. Galvanic cells
C. Electrolytic cells
D. Concentration cells

Correct answer: C

Rationale: The Structure → Function Reminder states that electrolytic cells use external power to force electrons against their thermodynamic preference.

MC 25. What is the oxidation number of Group 2 metals in compounds?

A. +1
B. +2
C. +3
D. 0

Correct answer: B

Rationale: According to the systematic rules provided, Group 2 metals have an oxidation number of +2 in compounds.

MC 26. In a polyatomic ion, what must the sum of all oxidation numbers equal?

A. Zero
B. The number of atoms
C. The ion's charge
D. +1

Correct answer: C

Rationale: The material states that in a polyatomic ion, the sum of all oxidation numbers must equal the ion's charge.

MC 27. What process occurs when an atom gains electrons?

A. Oxidation
B. Reduction
C. Ionization
D. Sublimation

Correct answer: B

Rationale: The material defines reduction as when an atom, ion, or molecule gains electrons, resulting in a decrease in its oxidation state.

MC 28. According to the material, what enables control over electron flow in electrochemical cells?

A. Temperature regulation

B. Pressure changes

C. Spatial separation of oxidation and reduction half-reactions

D. Magnetic fields

Correct answer: C

Rationale: The Structure → Function Reminder states that the spatial separation of oxidation and reduction half-reactions in electrochemical cells enables control over electron flow.

MC 29. What quantitative laws allow precise calculation of material and energy relationships in electrochemical processes?

A. Newton's laws

B. Faraday's laws

C. Boyle's laws

D. Charles' laws

Correct answer: B

Rationale: The Master Takeaways state that Faraday's laws allow precise calculation of material and energy relationships in electrochemical processes.

MC 30. What determines whether an electrochemical process stores or releases energy?

A. The temperature of the system

B. The concentration of reactants

C. The direction of electron flow between reduction potentials

D. The size of the electrodes

Correct answer: C

Rationale: The Master Takeaways state that the direction of electron flow between reduction potentials determines whether a process stores energy or releases energy.

MC 31. In the formation of NaCl, why does the reaction release significant energy?

A. The neutral atoms have higher potential energy than the ionic compound

B. Electrons are destroyed in the process

C. Sodium gains electrons from chlorine

D. The reaction absorbs heat from surroundings

Correct answer: A

Rationale: The material states that electron transfer in NaCl formation releases significant energy because the resulting ionic compound has much lower potential energy than the separated neutral atoms.

MC 32. What type of electrons participate in chemical bonding according to the material?

A. Core electrons

B. Valence electrons

C. All electrons equally

D. Neutrons

Correct answer: B

Rationale: The material states that valence electrons are those in the outermost shell that participate in chemical bonding.

MC 33. According to the Big Electrochemical Rule, what determines the driving force in electrochemical processes?

A. The mass of the electrodes

B. The difference in reduction potentials
C. The volume of the solution
D. The atmospheric pressure

Correct answer: B

Rationale: The Big Electrochemical Rule states that the driving force is determined by the difference in reduction potentials.

MC 34. What balancing method is mentioned for electrochemical processes?

A. Algebraic method
B. Inspection method
C. Half-reaction method
D. Matrix method

Correct answer: C

Rationale: The Master Takeaways mention systematic balancing methods using half-reactions for electrochemical processes.

MC 35. What do atoms typically try to resemble when gaining or losing electrons?

A. The nearest alkali metal
B. The nearest noble gas
C. The nearest halogen
D. The nearest transition metal

Correct answer: B

Rationale: The material states that atoms gain or lose valence electrons to achieve more stable electron configurations, typically resembling the nearest noble gas.

MC 36. In methane combustion products, what type of bonds contain the lower-energy electrons?

A. C-H and C-C bonds
B. C=C and C≡C bonds
C. C=O and O-H bonds
D. N-H and N=N bonds

Correct answer: C

Rationale: The material states that in methane combustion products, electrons are in lower-energy C=O and O-H bonds.

MC 37. What creates the electrical imbalances that power electrochemical cells?

A. Proton transfer between molecules
B. Electron transfer processes
C. Nuclear reactions
D. Magnetic field interactions

Correct answer: B

Rationale: The material states that electron transfer processes create the electrical imbalances that power electrochemical cells.

MC 38. According to the Final Diploma Alert, what should you always do when multiple reactions are possible?

A. Choose the fastest reaction
B. Compare reduction potentials systematically
C. Select the reaction with the most products
D. Pick the reaction with the highest temperature

Correct answer: B

Rationale: The Final Diploma Alert advises to always compare reduction potentials systematically when predicting products with multiple possible reactions.

MC 39. What depends on the actual reactions occurring, not just metal identities?

A. Electrode functions (anode/cathode)
B. Solution concentration
C. Temperature requirements
D. Pressure conditions

Correct answer: A

Rationale: The Final Diploma Alert states that electrode functions (anode/cathode) depend on the actual reactions occurring, not just the metal identities.

MC 40. What industrial applications rely on optimizing electron transfer principles?

A. Only battery technology
B. Only metal production
C. From battery technology to metal production
D. Only organic synthesis

Correct answer: C

Rationale: The Master Takeaways state that industrial applications from battery technology to metal production rely on optimizing fundamental electron transfer principles.

MC 41. What type of electron flow occurs in voltaic cells?

A. Forced flow requiring energy input
B. Spontaneous flow generating electricity
C. No electron flow
D. Random electron movement

Correct answer: B

Rationale: The Master Takeaways state that spontaneous electron flow in voltaic cells generates electricity.

MC 42. What enables chemical synthesis in electrolytic cells?

A. Spontaneous electron flow
B. Heat absorption
C. Forced electron flow
D. Natural diffusion

Correct answer: C

Rationale: The Master Takeaways state that forced electron flow in electrolytic cells enables chemical synthesis.

MC 43. According to the material, what makes combustion reactions essential for biological processes?

A. They produce water only
B. They consume oxygen
C. The energy difference appears as heat and light
D. They create new elements

Correct answer: C

Rationale: The material states that the energy difference in combustion appears as heat and light, making these reactions essential for biological processes in organisms.

MC 44. What process requires energy input according to the Master Takeaways?

A. Spontaneous processes
B. Non-spontaneous processes

C. All redox reactions
D. Only oxidation reactions

Correct answer: B

Rationale: The Master Takeaways indicate that non-spontaneous processes require energy input, as opposed to spontaneous processes that can do work.

MC 45. What provides the quantitative scale for predicting reaction spontaneity?

A. Oxidation numbers
B. Reduction potentials
C. Atomic masses
D. Bond energies

Correct answer: B

Rationale: The Master Takeaways state that reduction potentials provide the quantitative scale for predicting reaction spontaneity.

MC 46. In the reaction $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$, what happens to chlorine atoms?

A. Each loses two electrons
B. Each gains one electron
C. Each loses one electron
D. They remain neutral

Correct answer: B

Rationale: The material states that in NaCl formation, each chlorine atom gains one electron to form Cl^- ions.

MC 47. What type of process can do work according to the energy relationships discussed?

A. Non-spontaneous processes
B. Endothermic processes only
C. Spontaneous processes
D. All chemical processes equally

Correct answer: C

Rationale: The Master Takeaways indicate that spontaneous processes can do work, while non-spontaneous processes require energy input.

MC 48. According to the material, differences in what property between substances create reduction potential differences?

A. Atomic mass
B. Electron affinity
C. Melting point
D. Density

Correct answer: B

Rationale: The Cause → Effect Summary states that differences in electron affinity between substances create reduction potential differences.

MC 49. What can be harnessed in voltaic cells for power generation?

A. Heat flow
B. Electron flow
C. Proton flow
D. Light absorption

Correct answer: B

Rationale: The Cause → Effect Summary states that electron flow can be harnessed in voltaic cells for power generation.

MC 50. What type of calculation does the material mention for energy and material flows?

A. Approximate relationships

B. Qualitative relationships

C. Quantitative relationships

D. Random estimations

Correct answer: C

Rationale: The Cause → Effect Summary mentions that quantitative relationships (Faraday's laws, stoichiometry) allow precise calculation of energy and material flows.

MC 51. What systems exploit electrochemical principles for energy conversion according to the material?

A. Only industrial systems

B. Only biological systems

C. Industrial and biological systems

D. Only laboratory systems

Correct answer: C

Rationale: The Cause → Effect Summary states that industrial and biological systems exploit these principles for energy conversion.

MC 52. According to the material, what must happen when one species loses electrons through oxidation?

A. The electrons disappear

B. Another species must simultaneously gain those electrons

C. The electrons become protons

D. The reaction stops

Correct answer: B

Rationale: The material explains that when one species loses electrons through oxidation, another species must simultaneously gain those electrons through reduction.

MC 53. What is used for material production according to the Cause → Effect Summary?

A. Only thermal energy

B. Electrochemical principles

C. Nuclear reactions

D. Mechanical forces

Correct answer: B

Rationale: The Cause → Effect Summary states that industrial and biological systems exploit electrochemical principles for material production.

MC 54. What is the oxidation number of hydrogen in most compounds?

A. -1

B. 0

C. +1

D. +2

Correct answer: C

Rationale: According to the systematic rules, hydrogen has an oxidation number of +1 in compounds, except in metal hydrides where it's -1.

MC 55. What is the usual oxidation number of oxygen in compounds?

A. 0

B. -1
C. -2
D. +2

Correct answer: C

Rationale: The material states that oxygen has an oxidation number of -2 in compounds, except in peroxides where it's -1.

MC 56. According to the material, what can be driven in reverse in electrolytic cells?

A. Temperature changes
B. Pressure changes
C. Electron flow for chemical synthesis
D. Nuclear reactions

Correct answer: C

Rationale: The Cause → Effect Summary states that electron flow can be driven in reverse in electrolytic cells for chemical synthesis.

MC 57. What factor determines a system's efficiency according to the Final Transfer Prompt?

A. Only the electrode size
B. The color of the solution
C. How the same redox chemistry functions in both directions
D. The shape of the container

Correct answer: C

Rationale: The Final Transfer Prompt asks about factors determining system efficiency, focusing on how the same redox chemistry functions in both directions.

MC 58. What drives metabolic processes according to the material?

A. Only mechanical energy
B. Electrochemical principles
C. Solar radiation directly
D. Magnetic fields

Correct answer: B

Rationale: The Cause → Effect Summary indicates that biological systems exploit electrochemical principles for metabolic processes.

MC 59. What allows precise calculation according to the Master Takeaways?

A. Random sampling
B. Systematic balancing methods and quantitative laws
C. Estimation techniques
D. Visual observation

Correct answer: B

Rationale: The Master Takeaways state that systematic balancing methods (half-reactions) and quantitative laws (Faraday's laws) allow precise calculation.

MC 60. In electrochemical cells, what enables both power generation and chemical synthesis?

A. Temperature gradients
B. Controlled electron flow
C. Pressure differences
D. Light exposure

Correct answer: B

Rationale: The Master Takeaways emphasize that controlled electron flow enables both electricity generation in voltaic cells and chemical synthesis in electrolytic cells.

MC 61. According to the material, what compensates for the energy required to remove sodium's electron?

A. The ionization of chlorine

B. The lattice energy when Na^+ and Cl^- attract

C. The melting of sodium metal

D. The vaporization of chlorine

Correct answer: B

Rationale: The material states that the energy released when Na^+ and Cl^- ions attract in the crystal lattice (787 kJ/mol) more than compensates for the energy required to remove sodium's electron.

MC 62. What happens to an atom's oxidation state when it is reduced?

A. It increases

B. It stays the same

C. It decreases

D. It becomes neutral

Correct answer: C

Rationale: The material defines reduction as when an atom gains electrons, resulting in a decrease in its oxidation state.

MC 63. According to the material, what powers everything from car engines to cellular respiration?

A. Nuclear fission

B. The combustion of methane

C. Photosynthesis

D. Electromagnetic radiation

Correct answer: B

Rationale: The material specifically states that methane combustion powers everything from car engines to cellular respiration.

MC 64. What is created by spatial separation of redox half-reactions?

A. Heat generation only

B. Control over electron flow

C. Nuclear reactions

D. Chemical equilibrium

Correct answer: B

Rationale: The Structure → Function Reminder emphasizes that spatial separation of oxidation and reduction half-reactions enables control over electron flow.

MC 65. According to the Final Transfer Prompt, what should an electrochemical system be able to do?

A. Only store electrical energy

B. Only release electrical energy

C. Both store and release electrical energy

D. Convert matter to energy

Correct answer: C

Rationale: The Final Transfer Prompt asks to design a system that could both store electrical energy when available and release it when needed.

MC 66. What determines practical viability according to the Final Transfer Prompt?

A. Color changes in the system

B. Factors affecting system efficiency
C. The size of the container
D. The age of the electrodes

Correct answer: B

Rationale: The Final Transfer Prompt mentions that factors determining the system's efficiency affect its practical viability.

MC 67. In diploma exam questions about non-standard conditions, what should you do according to the material?

A. Ignore the conditions
B. Compare reduction potentials systematically
C. Use standard conditions only
D. Estimate the answer

Correct answer: B

Rationale: The Final Diploma Alert advises to always compare reduction potentials systematically when dealing with non-standard conditions.

MC 68. What fundamental principle drives all redox and electrochemical processes?

A. Proton transfer
B. Electron transfer
C. Heat transfer
D. Mass transfer

Correct answer: B

Rationale: The Master Takeaways state that electron transfer drives all redox and electrochemical processes.

MC 69. Industrial applications optimize electron transfer principles under what conditions?

A. Only standard conditions
B. Different conditions and constraints
C. Vacuum conditions only
D. High pressure only

Correct answer: B

Rationale: The Master Takeaways state that industrial applications rely on optimizing fundamental electron transfer principles under different conditions and constraints.

MC 70. What happens during the formation of ionic compounds?

A. Electrons are shared equally
B. Electron transfer drives the formation
C. Protons are exchanged
D. Neutrons are transferred

Correct answer: B

Rationale: The material states that electron transfer process drives the formation of ionic compounds.

MC 71. According to the material, what energy levels do electrons occupy?

A. Random energy levels
B. Specific energy levels around the nucleus
C. Only one energy level
D. Continuously variable levels

Correct answer: B

Rationale: The material states that electrons exist in specific energy levels around the nucleus.

MC 72. What shell contains the valence electrons?

A. The innermost shell
B. The middle shell
C. The outermost shell
D. All shells equally

Correct answer: C

Rationale: The material defines valence electrons as those in the outermost shell.

MC 73. What is another name for oxidation-reduction reactions?

A. Acid-base reactions
B. Redox reactions
C. Precipitation reactions
D. Substitution reactions

Correct answer: B

Rationale: The material states that oxidation-reduction reactions are also called redox reactions.

MC 74. According to the material, what happens to electrons during oxidation?

A. They are gained
B. They are lost
C. They remain unchanged
D. They are created

Correct answer: B

Rationale: The material defines oxidation as when an atom, ion, or molecule loses electrons.

MC 75. What completes chlorine's valence shell when it gains an electron?

A. A duplet configuration
B. A sextet configuration
C. An octet configuration
D. A decet configuration

Correct answer: C

Rationale: The material states that adding an electron to chlorine completes its valence shell, creating a stable octet configuration.

Unit B Review

Constructed Response Bank

Question 1

A student claims that in the reaction $2Na + Cl_2 \rightarrow 2NaCl$, sodium is reduced because it forms a positive ion.

a. State whether the student's claim is correct or incorrect and explain what actually happens to sodium in terms of electron transfer and oxidation state change.

b. Analyze the energy changes in this reaction by explaining why the formation of NaCl releases energy despite requiring energy to remove an electron from sodium.

c. Identify one practical application where this type of electron transfer reaction is utilized.

Question 2

During methane combustion ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$), carbon undergoes a dramatic change in oxidation state.

a. Calculate the oxidation number of carbon in CH_4 and in CO_2, showing your systematic approach.

b. Explain how the change in oxidation number relates to electron transfer and identify how many electrons are lost per carbon atom.

c. State one reason why this electron transfer releases energy that powers biological processes.

Question 3

A researcher observes that chlorine has a high electron affinity of 349 kJ/mol while sodium has a low ionization energy of 496 kJ/mol.

a. Explain why sodium has such a low ionization energy by referring to its electronic structure.

b. Analyze why chlorine readily accepts electrons by explaining what electron configuration it achieves.

c. Identify which species is oxidized and which is reduced when sodium reacts with chlorine.

Question 4

An electrochemistry student states that oxidation and reduction can occur independently in separate reactions.

a. Evaluate this statement and explain why oxidation and reduction must occur together.

b. Explain how electrochemical cells utilize the spatial separation of these coupled processes.

c. Name one type of cell that generates electricity from spontaneous redox reactions.

Question 5

A chemistry teacher demonstrates that elements in their free state always have an oxidation number of zero.

a. Explain why free elements like Na, Cl_2, and O_2 all have oxidation numbers of zero.

b. Compare the oxidation number rules for Group 1 and Group 2 metals in compounds and explain the pattern.

c. State the oxidation number of fluorine in all its compounds.

Question 6

Research shows that the thermodynamic driving force for redox reactions involves electron movement between different energy levels.

a. Describe the direction of electron movement that provides the thermodynamic driving force for redox reactions.

b. Analyze how this principle applies to methane combustion by comparing the bond energies involved.

c. Name one practical use of the energy released from methane combustion.

Question 7

In a polyatomic ion, the sum of oxidation numbers follows a specific rule different from neutral compounds.

a. State the rule for the sum of oxidation numbers in neutral compounds versus polyatomic ions.

b. Explain how to use these rules to determine unknown oxidation numbers in compounds.

c. State the oxidation number of hydrogen in metal hydrides.

Question 8

An experiment shows that NaCl formation releases 787 kJ/mol as lattice energy when the ions come together.

a. Define lattice energy and explain why it is released when Na^+ and Cl^- ions form a crystal.

b. Analyze the overall energy balance by comparing the lattice energy to the energy required for electron transfer.

c. Identify what type of compound forms when electron transfer creates ions.

Question 9

A student observes that reduction potentials provide a quantitative scale for predicting reaction spontaneity.

a. Explain what reduction potentials measure and how they predict spontaneity.

b. Describe the relationship between reduction potential differences and the driving force for electron transfer.

c. State the direction of spontaneous electron flow in terms of reduction potentials.

Question 10

Industrial processes use electrolytic cells to force electrons against their thermodynamic preference.

a. Explain what it means to force electrons against their thermodynamic preference and why external power is needed.

b. Compare how voltaic and electrolytic cells differ in their use of electron flow.

c. Name one industrial application that uses electrolytic cells.

Question 11

Faraday's laws allow precise calculation of material and energy relationships in electrochemical processes.

a. Explain the significance of Faraday's laws in electrochemistry and what they allow us to calculate.

b. Describe how stoichiometry combines with Faraday's laws to predict electrochemical outcomes.

c. Identify what balancing method is used for electrochemical processes.

Question 12

A researcher notes that biological systems exploit electrochemical principles for metabolic processes.

a. Explain how cellular respiration relates to the combustion of organic molecules like methane.

b. Analyze why electron transfer reactions are essential for biological energy conversion.

c. Name one biological process powered by the energy from redox reactions.

Question 13

An electrochemical system can both store electrical energy when power is available and release it when needed.

a. Explain how the same redox chemistry can function in both directions for energy storage and release.

b. Analyze what factors determine the efficiency of such an electrochemical storage system.

c. Name one example of a device that stores and releases electrical energy electrochemically.

Question 14

Exam questions often present non-standard conditions for electrochemical cells.

a. Explain why comparing reduction potentials systematically is crucial when predicting products in electrochemical reactions.

b. Describe how electrode functions depend on actual reactions rather than just metal identities.

c. State what determines whether a process stores or releases energy.

Question 15

A student notices that oxygen has different oxidation numbers in different compounds.

a. State the typical oxidation number of oxygen in compounds and identify the exception with its oxidation number.

b. Explain why oxygen in O_2 has an oxidation number of zero while in H_2O it has -2.

c. Name the process that occurs when O_2 becomes H_2O in combustion reactions.

Question 16

Industrial applications optimize electron transfer principles under different conditions and constraints.

a. Explain why industrial processes must optimize electron transfer under various conditions rather than just standard conditions.

b. Describe how both battery technology and metal production rely on the same fundamental electron transfer principles.

c. Identify whether metal production typically uses voltaic or electrolytic cells.

Question 17

Valence electrons in atoms determine their chemical bonding behavior.

a. Define valence electrons and explain their role in oxidation-reduction reactions.

b. Analyze why atoms tend to achieve noble gas configurations through electron transfer.

c. State what type of compound forms when atoms achieve stable configurations through complete electron transfer.

Question 18

The combustion of methane powers various systems from car engines to cellular respiration.

a. Describe the electron transfer that occurs during methane combustion in terms of the oxidation states of carbon.

b. Explain why this reaction releases enough energy to power both mechanical and biological systems.

c. Name one specific application where methane combustion provides energy.

Question 19

A student wonders why some redox reactions occur spontaneously while others require energy input.

a. Explain what determines whether a redox reaction is spontaneous or non-spontaneous.

b. Compare how spontaneous and non-spontaneous processes are utilized in different types of electrochemical cells.

c. Identify which type of process can do work.

Question 20

The spatial separation of half-reactions is fundamental to electrochemical cell operation.

a. Explain why spatial separation of oxidation and reduction half-reactions is necessary for electrochemical cells to function.

b. Describe how this separation enables different functions in voltaic versus electrolytic cells.

c. State what connects the separated half-cells to complete the circuit.

Model Answers & Rubrics

Question 1

Part a: State whether the student's claim is correct or incorrect and explain what actually happens to sodium in terms of electron transfer and oxidation state change.

Model Answer: The student's claim is incorrect. Sodium is oxidized, not reduced, in this reaction. Sodium loses one electron (going from Na to Na^+), which is the definition of oxidation. Its oxidation number increases from 0 to +1, confirming oxidation has occurred. (2 marks: 1 for identifying incorrect, 1 for explaining oxidation with electron loss and oxidation number increase) (2 marks)

Part b: Analyze the energy changes in this reaction by explaining why the formation of NaCl releases energy despite requiring energy to remove an electron from sodium.

Model Answer: Although removing an electron from sodium requires 496 kJ/mol (ionization energy), the overall reaction releases energy because: chlorine's high electron affinity (349 kJ/mol) partially compensates for this, and more importantly, the lattice energy (787 kJ/mol) released when Na^+ and Cl^- ions attract in the crystal lattice more than compensates for the energy input, making the net reaction highly exothermic. (2 marks: 1 for identifying energy requirements, 1 for explaining lattice energy compensation) (2 marks)

Part c: Identify one practical application where this type of electron transfer reaction is utilized.

Model Answer: The formation of table salt (NaCl) in industrial processes, or the use of sodium compounds in chemical synthesis where controlled oxidation of sodium is required. (1 mark for any valid practical application) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for correctly identifying the claim as incorrect, 1 mark for explaining oxidation with electron loss and oxidation state change

Part b (2 marks): 1 mark for discussing energy requirements (ionization energy), 1 mark for explaining lattice energy compensation

Part c (1 mark): 1 mark for providing a valid practical application

Question 2

Part a: Calculate the oxidation number of carbon in CH_4 and in CO_2, showing your systematic approach.

Model Answer: In CH_4: H has oxidation number +1, with 4 H atoms giving +4 total. For a neutral compound, C must be -4. In CO_2: O has oxidation number -2, with 2 O atoms giving -4 total. For a neutral compound, C must be +4. (2 marks: 1 for correct calculation in CH_4, 1 for correct calculation in CO_2) (2 marks)

Part b: Explain how the change in oxidation number relates to electron transfer and identify how many electrons are lost per carbon atom.

Model Answer: Carbon's oxidation number increases from -4 to +4, an increase of 8. This means carbon loses 8 electrons per atom during the reaction. This electron loss represents oxidation of carbon, while oxygen gains these electrons and is reduced from 0 to -2. (2 marks: 1 for identifying 8 electron loss, 1 for connecting to oxidation/reduction) (2 marks)

Part c: State one reason why this electron transfer releases energy that powers biological processes.

Model Answer: The electrons move from high-energy C-H bonds to lower-energy C=O and O-H bonds, and this energy difference is released as heat and light. (1 mark for identifying bond energy difference) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for CH_4 calculation showing C = -4, 1 mark for CO_2 calculation showing C = +4

Part b (2 marks): 1 mark for identifying 8 electron loss, 1 mark for explaining oxidation/reduction relationship

Part c (1 mark): 1 mark for identifying energy difference between bond types

Question 3

Part a: Explain why sodium has such a low ionization energy by referring to its electronic structure.

Model Answer: Sodium has a low ionization energy because its single valence electron is far from the nucleus and is shielded by inner electron shells. This shielding effect reduces the attractive force between the nucleus and the valence electron, making it easier to remove. (2 marks: 1 for distance from nucleus, 1 for shielding effect) (2 marks)

Part b: Analyze why chlorine readily accepts electrons by explaining what electron configuration it achieves.

Model Answer: Chlorine readily accepts electrons because gaining one electron completes its valence shell, creating a stable octet configuration. This octet represents the electron configuration of the nearest noble gas (argon), which is particularly stable due to having a full outer shell. (2 marks: 1 for octet completion, 1 for noble gas configuration stability) (2 marks)

Part c: Identify which species is oxidized and which is reduced when sodium reacts with chlorine.

Model Answer: Sodium is oxidized (loses electrons) and chlorine is reduced (gains electrons). (1 mark for correctly identifying both) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for distance from nucleus, 1 mark for shielding effect

Part b (2 marks): 1 mark for octet completion, 1 mark for noble gas stability

Part c (1 mark): 1 mark for correctly identifying sodium as oxidized and chlorine as reduced

Question 4

Part a: Evaluate this statement and explain why oxidation and reduction must occur together.

Model Answer: The statement is incorrect. Oxidation and reduction must occur together because electrons cannot exist freely in solution. When one species loses electrons through oxidation, another species must simultaneously gain those electrons through reduction. This coupling is why they're called redox reactions. (2 marks: 1 for identifying incorrect, 1 for explaining electron coupling) (2 marks)

Part b: Explain how electrochemical cells utilize the spatial separation of these coupled processes.

Model Answer: Electrochemical cells spatially separate the oxidation and reduction half-reactions while maintaining electrical connection. This separation enables control over electron flow - voltaic cells force electrons through external circuits to generate power, while electrolytic cells use external power to force electrons against their thermodynamic preference. (2 marks: 1 for spatial separation concept, 1 for control mechanism) (2 marks)

Part c: Name one type of cell that generates electricity from spontaneous redox reactions.

Model Answer: Voltaic cells (or galvanic cells) generate electricity from spontaneous redox reactions. (1 mark for correct identification) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying incorrect, 1 mark for explaining electron coupling requirement

Part b (2 marks): 1 mark for spatial separation, 1 mark for control mechanism

Part c (1 mark): 1 mark for identifying voltaic cells

Question 5

Part a: Explain why free elements like Na, Cl_2, and O_2 all have oxidation numbers of zero.

Model Answer: Free elements have oxidation number zero because they haven't gained or lost electrons - they exist in their elemental, uncharged state. In diatomic molecules like Cl_2 and O_2, the electrons are shared equally between identical atoms, so neither atom has gained or lost electrons relative to its elemental state. (2 marks: 1 for uncharged state, 1 for equal sharing in diatomics) (2 marks)

Part b: Compare the oxidation number rules for Group 1 and Group 2 metals in compounds and explain the pattern.

Model Answer: Group 1 metals always have oxidation number +1 in compounds, while Group 2 metals have +2. This pattern reflects their valence electron configurations - Group 1 has one valence electron to lose, Group 2 has two valence electrons to lose, both achieving noble gas configurations. (2 marks: 1 for stating rules correctly, 1 for explaining valence electron basis) (2 marks)

Part c: State the oxidation number of fluorine in all its compounds.

Model Answer: Fluorine has an oxidation number of -1 in all its compounds. (1 mark for correct answer) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for elemental state explanation, 1 mark for equal sharing in diatomics

Part b (2 marks): 1 mark for correct rules, 1 mark for valence electron explanation

Part c (1 mark): 1 mark for -1

Question 6

Part a: Describe the direction of electron movement that provides the thermodynamic driving force for redox reactions.

Model Answer: The thermodynamic driving force comes from electrons moving from higher-energy orbitals to lower-energy orbitals. This movement to lower energy states releases energy, making the reaction thermodynamically favorable. (2 marks: 1 for identifying high to low movement, 1 for energy release connection) (2 marks)

Part b: Analyze how this principle applies to methane combustion by comparing the bond energies involved.

Model Answer: In methane combustion, electrons move from high-energy C-H bonds in CH_4 to lower-energy C=O bonds in CO_2 and O-H bonds in H_2O. This transition from higher to lower energy bonds releases the energy difference as heat and light, driving the reaction forward. (2 marks: 1 for identifying bond types, 1 for energy difference explanation) (2 marks)

Part c: Name one practical use of the energy released from methane combustion.

Model Answer: Heating homes, generating electricity in power plants, or powering car engines. (1 mark for any valid application) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for high to low energy movement, 1 mark for energy release

Part b (2 marks): 1 mark for bond type comparison, 1 mark for energy difference

Part c (1 mark): 1 mark for valid practical application

Question 7

Part a: State the rule for the sum of oxidation numbers in neutral compounds versus polyatomic ions.

Model Answer: In neutral compounds, the sum of all oxidation numbers must equal zero. In polyatomic ions, the sum of all oxidation numbers must equal the ion's charge. (2 marks: 1 for neutral compound rule, 1 for polyatomic ion rule) (2 marks)

Part b: Explain how to use these rules to determine unknown oxidation numbers in compounds.

Model Answer: To find an unknown oxidation number: first assign known oxidation numbers using the systematic rules (Group 1 = +1, Group 2 = +2, O = -2, etc.), then set up an equation where the sum equals zero (neutral) or the ion charge (polyatomic), and solve for the unknown. (2 marks: 1 for systematic assignment, 1 for equation setup and solving) (2 marks)

Part c: State the oxidation number of hydrogen in metal hydrides.

Model Answer: In metal hydrides, hydrogen has an oxidation number of -1. (1 mark for correct answer) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for neutral compound rule, 1 mark for polyatomic ion rule

Part b (2 marks): 1 mark for systematic assignment, 1 mark for equation method

Part c (1 mark): 1 mark for -1

Question 8

Part a: Define lattice energy and explain why it is released when Na^+ and Cl^- ions form a crystal.

Model Answer: Lattice energy is the energy released when gaseous ions combine to form a crystalline solid. It's released because the electrostatic attraction between oppositely charged Na^+ and Cl^- ions in the crystal lattice results in a much lower potential energy state than separated ions. (2 marks: 1 for definition, 1 for electrostatic attraction explanation) (2 marks)

Part b: Analyze the overall energy balance by comparing the lattice energy to the energy required for electron transfer.

Model Answer: The ionization energy to remove an electron from Na is 496 kJ/mol, partially offset by Cl's electron affinity of 349 kJ/mol, giving a net energy requirement of 147 kJ/mol for electron transfer. The lattice energy of 787 kJ/mol more than compensates for this, resulting in a net energy release of 640 kJ/mol. (2 marks: 1 for calculating electron transfer energy, 1 for net energy analysis) (2 marks)

Part c: Identify what type of compound forms when electron transfer creates ions.

Model Answer: Ionic compounds form when electron transfer creates ions. (1 mark for correct identification) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for lattice energy definition, 1 mark for electrostatic explanation

Part b (2 marks): 1 mark for electron transfer calculation, 1 mark for net energy balance

Part c (1 mark): 1 mark for ionic compounds

Question 9

Part a: Explain what reduction potentials measure and how they predict spontaneity.

Model Answer: Reduction potentials measure the tendency of a species to gain electrons (be reduced). They predict spontaneity because electron transfer occurs spontaneously from species with lower reduction potential to those with higher reduction potential - the greater the difference, the more spontaneous the reaction. (2 marks: 1 for definition, 1 for spontaneity prediction) (2 marks)

Part b: Describe the relationship between reduction potential differences and the driving force for electron transfer.

Model Answer: The difference in reduction potentials between two species determines the driving force - larger differences create stronger driving forces for electron transfer. This difference also determines the voltage in electrochemical cells and indicates how much energy can be harnessed or must be supplied. (2 marks: 1 for difference-force relationship, 1 for energy/voltage connection) (2 marks)

Part c: State the direction of spontaneous electron flow in terms of reduction potentials.

Model Answer: Electrons flow spontaneously from lower to higher reduction potential. (1 mark for correct direction) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for reduction potential definition, 1 mark for spontaneity prediction

Part b (2 marks): 1 mark for difference-force relationship, 1 mark for energy implications

Part c (1 mark): 1 mark for correct flow direction

Question 10

Part a: Explain what it means to force electrons against their thermodynamic preference and why external power is needed.

Model Answer: Forcing electrons against thermodynamic preference means driving them from higher to lower reduction potential, opposite to spontaneous flow. External power is needed because this non-spontaneous process requires energy input to overcome the natural tendency of electrons to flow the other way. (2 marks: 1 for direction explanation, 1 for energy requirement) (2 marks)

Part b: Compare how voltaic and electrolytic cells differ in their use of electron flow.

Model Answer: Voltaic cells harness spontaneous electron flow (from lower to higher reduction potential) to generate electricity. Electrolytic cells use external electricity to force non-spontaneous electron flow (from higher to lower reduction potential) to drive chemical synthesis or decomposition reactions. (2 marks: 1 for voltaic description, 1 for electrolytic description) (2 marks)

Part c: Name one industrial application that uses electrolytic cells.

Model Answer: Metal production (like aluminum extraction), electroplating, or water electrolysis for hydrogen production. (1 mark for valid application) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for non-spontaneous direction, 1 mark for energy input requirement

Part b (2 marks): 1 mark for voltaic cell description, 1 mark for electrolytic cell description

Part c (1 mark): 1 mark for valid industrial application

Question 11

Part a: Explain the significance of Faraday's laws in electrochemistry and what they allow us to calculate.

Model Answer: Faraday's laws relate the amount of electrical charge passed through an electrochemical cell to the amount of chemical change. They allow precise calculation of how much material is produced or consumed based on the current and time, or how much electrical energy is needed for a specific chemical transformation. (2 marks: 1 for charge-chemistry relationship, 1 for calculation applications) (2 marks)

Part b: Describe how stoichiometry combines with Faraday's laws to predict electrochemical outcomes.

Model Answer: Stoichiometry determines the mole ratios of reactants and products in the redox reaction, while Faraday's laws connect moles of electrons transferred to electrical charge. Together, they allow calculation of exact amounts of products formed or reactants consumed for a given amount of electrical current. (2 marks: 1 for stoichiometry role, 1 for integration with Faraday's laws) (2 marks)

Part c: Identify what balancing method is used for electrochemical processes.

Model Answer: The half-reaction method is used for balancing electrochemical processes. (1 mark for correct method) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for charge-chemistry relationship, 1 mark for calculation capabilities

Part b (2 marks): 1 mark for stoichiometry role, 1 mark for integration explanation

Part c (1 mark): 1 mark for half-reaction method

Question 12

Part a: Explain how cellular respiration relates to the combustion of organic molecules like methane.

Model Answer: Cellular respiration is essentially controlled combustion of organic molecules. Like methane combustion, it involves oxidation of carbon-hydrogen bonds and reduction of oxygen, releasing energy from high-energy C-H bonds to form lower-energy C=O and O-H bonds. The energy is captured in biological systems rather than released as heat and light. (2 marks: 1 for oxidation-reduction parallel, 1 for energy capture difference) (2 marks)

Part b: Analyze why electron transfer reactions are essential for biological energy conversion.

Model Answer: Electron transfer reactions allow organisms to extract energy from food molecules in controlled steps. The movement of electrons from higher to lower energy states releases energy that can be captured in ATP or other energy carriers, enabling life processes. Without these redox reactions, organisms couldn't access chemical energy. (2 marks: 1 for controlled energy extraction, 1 for biological energy storage) (2 marks)

Part c: Name one biological process powered by the energy from redox reactions.

Model Answer: ATP synthesis, muscle contraction, or active transport across membranes. (1 mark for valid biological process) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for combustion parallel, 1 mark for energy capture difference

Part b (2 marks): 1 mark for controlled extraction, 1 mark for energy storage mechanism

Part c (1 mark): 1 mark for valid biological process

Question 13

Part a: Explain how the same redox chemistry can function in both directions for energy storage and release.

Model Answer: The same redox reaction can be reversible - running spontaneously in one direction (voltaic mode) to release energy, or forced in reverse (electrolytic mode) to store energy. During charging, external power forces electrons against their preference, storing energy in chemical bonds. During discharge, electrons flow spontaneously, releasing stored energy. (2 marks: 1 for reversibility concept, 1 for charge/discharge explanation) (2 marks)

Part b: Analyze what factors determine the efficiency of such an electrochemical storage system.

Model Answer: Efficiency depends on: the reduction potential difference (determining voltage and energy density), the reversibility of the electrode reactions (some energy lost to side reactions), internal resistance causing energy loss as heat, and the stability of electrode materials over multiple cycles. (2 marks: 1 for potential/reversibility factors, 1 for resistance/stability factors) (2 marks)

Part c: Name one example of a device that stores and releases electrical energy electrochemically.

Model Answer: Rechargeable batteries (lithium-ion, lead-acid, etc.) or fuel cells with reversible operation. (1 mark for valid example) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for reversibility, 1 mark for charge/discharge mechanism

Part b (2 marks): 1 mark for thermodynamic factors, 1 mark for practical factors

Part c (1 mark): 1 mark for valid device example

Question 14

Part a: Explain why comparing reduction potentials systematically is crucial when predicting products in electrochemical reactions.

Model Answer: Systematic comparison of reduction potentials determines which species will be oxidized and which reduced - the species with higher reduction potential gets reduced while the one with lower potential gets oxidized. When multiple reactions are possible, the pair with the largest potential difference occurs preferentially. (2 marks: 1 for oxidation/reduction determination, 1 for selecting among multiple possibilities) (2 marks)

Part b: Describe how electrode functions depend on actual reactions rather than just metal identities.

Model Answer: The anode is always where oxidation occurs and the cathode where reduction occurs, regardless of the metal present. In voltaic cells, the metal with lower reduction potential becomes the anode. In electrolytic cells, external voltage can force any electrode to be anode or cathode depending on the applied polarity. (2 marks: 1 for reaction-based definition, 1 for cell type differences) (2 marks)

Part c: State what determines whether a process stores or releases energy.

Model Answer: The direction of electron flow between reduction potentials determines this - spontaneous flow releases energy, forced flow stores energy. (1 mark for correct relationship) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for potential comparison method, 1 mark for selecting preferred reaction

Part b (2 marks): 1 mark for reaction-based electrode definition, 1 mark for cell type variation

Part c (1 mark): 1 mark for flow direction determining energy storage/release

Question 15

Part a: State the typical oxidation number of oxygen in compounds and identify the exception with its oxidation number.

Model Answer: Oxygen typically has an oxidation number of -2 in compounds. The exception is in peroxides, where oxygen has an oxidation number of -1. (2 marks: 1 for typical -2, 1 for peroxide exception -1) (2 marks)

Part b: Explain why oxygen in O_2 has an oxidation number of zero while in H_2O it has -2.

Model Answer: In O_2, oxygen atoms share electrons equally since they're identical, so neither gains nor loses electrons relative to the elemental state (oxidation number = 0). In H_2O, oxygen is more electronegative than hydrogen and effectively gains 2 electrons, giving it an oxidation number of -2. (2 marks: 1 for O_2 equal sharing, 1 for H_2O electron gain) (2 marks)

Part c: Name the process that occurs when O_2 becomes H_2O in combustion reactions.

Model Answer: Reduction occurs (oxygen gains electrons, oxidation number decreases from 0 to -2). (1 mark for identifying reduction) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for -2 typical, 1 mark for -1 in peroxides

Part b (2 marks): 1 mark for O_2 explanation, 1 mark for H_2O explanation

Part c (1 mark): 1 mark for reduction

Question 16

Part a: Explain why industrial processes must optimize electron transfer under various conditions rather than just standard conditions.

Model Answer: Industrial processes face real-world constraints like temperature variations, pressure requirements, economic factors, and material availability. Optimizing under these conditions ensures maximum efficiency, product yield, and cost-effectiveness while maintaining safety and environmental standards. Standard conditions rarely exist in industrial settings. (2 marks: 1 for practical constraints, 1 for optimization benefits) (2 marks)

Part b: Describe how both battery technology and metal production rely on the same fundamental electron transfer principles.

Model Answer: Battery technology uses controlled electron transfer between species with different reduction potentials to store/release energy. Metal production uses electrolysis to force electron transfer, reducing metal ions to pure metal. Both rely on managing reduction potentials and controlling electron flow, just in opposite directions for different purposes. (2 marks: 1 for battery principle, 1 for metal production principle) (2 marks)

Part c: Identify whether metal production typically uses voltaic or electrolytic cells.

Model Answer: Metal production typically uses electrolytic cells (to force the reduction of metal ions). (1 mark for electrolytic cells) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying constraints, 1 mark for optimization rationale

Part b (2 marks): 1 mark for battery technology, 1 mark for metal production

Part c (1 mark): 1 mark for electrolytic cells

Question 17

Part a: Define valence electrons and explain their role in oxidation-reduction reactions.

Model Answer: Valence electrons are those in the outermost shell of an atom. In redox reactions, these are the electrons that are transferred - atoms lose valence electrons during oxidation or gain them during reduction to achieve more stable electron configurations, typically resembling noble gases. (2 marks: 1 for definition, 1 for role in redox) (2 marks)

Part b: Analyze why atoms tend to achieve noble gas configurations through electron transfer.

Model Answer: Noble gas configurations represent filled valence shells (usually octets), which are particularly stable due to maximum electron-electron pairing and symmetrical electron distribution. Atoms gain or lose electrons to achieve these stable configurations because they represent energy minima - the most stable, lowest energy arrangements. (2 marks: 1 for filled shell stability, 1 for energy minimization) (2 marks)

Part c: State what type of compound forms when atoms achieve stable configurations through complete electron transfer.

Model Answer: Ionic compounds form when atoms achieve stable configurations through complete electron transfer. (1 mark for ionic compounds) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for valence electron definition, 1 mark for transfer role

Part b (2 marks): 1 mark for stability explanation, 1 mark for energy consideration

Part c (1 mark): 1 mark for ionic compounds

Question 18

Part a: Describe the electron transfer that occurs during methane combustion in terms of the oxidation states of carbon.

Model Answer: During methane combustion, carbon is oxidized from -4 in CH_4 to +4 in CO_2, losing 8 electrons per carbon atom. These electrons are transferred to oxygen atoms, which are reduced from 0 in O_2 to -2 in the products (CO_2 and H_2O). (2 marks: 1 for carbon oxidation description, 1 for oxygen reduction) (2 marks)

Part b: Explain why this reaction releases enough energy to power both mechanical and biological systems.

Model Answer: The reaction releases enormous energy because electrons move from high-energy C-H bonds to much lower-energy C=O and O-H bonds. This large energy difference between reactant and product bonds provides sufficient energy for mechanical work in engines and for capturing in ATP molecules during cellular respiration. (2 marks: 1 for bond energy difference, 1 for application to both systems) (2 marks)

Part c: Name one specific application where methane combustion provides energy.

Model Answer: Heating homes, generating electricity in power plants, or providing energy for cellular respiration. (1 mark for any valid application) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for carbon oxidation details, 1 mark for oxygen reduction

Part b (2 marks): 1 mark for bond energy explanation, 1 mark for dual application

Part c (1 mark): 1 mark for specific application

Question 19

Part a: Explain what determines whether a redox reaction is spontaneous or non-spontaneous.

Model Answer: The difference in reduction potentials determines spontaneity. If electron flow goes from lower to higher reduction potential (following the natural gradient), the reaction is spontaneous and releases energy. If forcing electrons from higher to lower potential (against the gradient), the reaction is non-spontaneous and requires energy input. (2 marks: 1 for reduction potential difference, 1 for flow direction) (2 marks)

Part b: Compare how spontaneous and non-spontaneous processes are utilized in different types of electrochemical cells.

Model Answer: Spontaneous processes are used in voltaic cells where the natural electron flow generates electrical energy that can do work. Non-spontaneous processes occur in electrolytic cells where external electrical energy forces unfavorable reactions to proceed, enabling energy storage or chemical synthesis. (2 marks: 1 for voltaic/spontaneous, 1 for electrolytic/non-spontaneous) (2 marks)

Part c: Identify which type of process can do work.

Model Answer: Spontaneous processes can do work (release usable energy). (1 mark for spontaneous) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for reduction potential role, 1 mark for direction determining spontaneity

Part b (2 marks): 1 mark for voltaic cell description, 1 mark for electrolytic cell description

Part c (1 mark): 1 mark for spontaneous processes

Question 20

Part a: Explain why spatial separation of oxidation and reduction half-reactions is necessary for electrochemical cells to function.

Model Answer: Spatial separation prevents direct electron transfer between reactants, forcing electrons to travel through an external circuit. This controlled electron flow through the wire generates electrical current that can do useful work or be controlled by external voltage, which wouldn't be possible if the reactions occurred in the same location. (2 marks: 1 for preventing direct transfer, 1 for enabling controlled flow) (2 marks)

Part b: Describe how this separation enables different functions in voltaic versus electrolytic cells.

Model Answer: In voltaic cells, separation allows spontaneous electron flow through external circuits to generate usable electricity. In electrolytic cells, separation allows external power sources to force electrons through the circuit in the non-spontaneous direction, driving chemical reactions that wouldn't occur naturally. The same separation principle enables opposite functions. (2 marks: 1 for voltaic function, 1 for electrolytic function) (2 marks)

Part c: State what connects the separated half-cells to complete the circuit.

Model Answer: A salt bridge or porous membrane connects the half-cells to allow ion flow and maintain electrical neutrality. (1 mark for salt bridge/membrane) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for preventing direct transfer, 1 mark for controlled flow benefit

Part b (2 marks): 1 mark for voltaic cell function, 1 mark for electrolytic cell function

Part c (1 mark): 1 mark for salt bridge or equivalent